

Epidemic Risk Intelligence Report

Chennai Metro General Hospital

Chennai Urban District · Census Code: IN-TN-CHN-042 · Generated: 2026-06-22 13:40:42 UTC+5:30

Key Performance Indicators

78.4%

Outbreak Probability

142

Predicted Cases (7-day)

18

EHR Records Uploaded

Active Alert Status

▲ ALERT LEVEL: HIGH — Outbreak Probability 78.4% | Predicted Cases: 142

Immediate epidemiological review recommended based on current sensor and EHR data fusion.

EHR Patient Summary

18

Total Records

6

Dengue Positive

12

Dengue Negative

33.3%

Positivity Rate

37.8°C

Avg Temp

Positivity Rate: 33.3%

33.3%

File / Record ID	Temperature (°C)	Dengue Status
EHR_TN_1000.json	37.2	Positive
EHR_TN_1001.json	37.5	Negative
EHR_TN_1002.json	37.8	Negative
EHR_TN_1003.json	38.1	Positive
EHR_TN_1004.json	38.4	Negative
EHR_TN_1005.json	37.2	Negative
EHR_TN_1006.json	37.5	Positive
EHR_TN_1007.json	37.8	Negative
EHR_TN_1008.json	38.1	Negative

EHR_TN_1009.json	38.4	Positive
EHR_TN_1010.json	37.2	Negative
EHR_TN_1011.json	37.5	Negative
EHR_TN_1012.json	37.8	Positive
EHR_TN_1013.json	38.1	Negative
EHR_TN_1014.json	38.4	Negative
EHR_TN_1015.json	37.2	Positive
EHR_TN_1016.json	37.5	Negative
EHR_TN_1017.json	37.8	Negative

SHAP Feature Importance

Features ranked by absolute SHAP value contribution to the outbreak risk prediction. Red bars = risk-increasing features; green bars = risk-reducing features.

Rank	Feature Name	SHAP Score	Attribution
1	PM2.5 Concentration	+0.3812	100.0%
2	Ambient Humidity (%)	+0.2945	77.3%
3	Stagnant Water Index	+0.2301	60.4%
4	Dengue Case History 30d	+0.1987	52.1%
5	Average Temperature	+0.1654	43.4%
6	CO2 Level (ppm)	+0.0923	24.2%
7	Rainfall (mm)	+0.0811	21.3%
8	Population Density	+0.0702	18.4%
9	Vaccination Coverage	-0.1234	32.4%
10	Wind Speed	-0.0654	17.2%

Epidemiological Timeline

Week	Year	Actual Cases	Predicted Cases	Probability %	Risk Level
20	2025	18	21	31.0%	LOW
21	2025	27	24	42.0%	MEDIUM
22	2025	35	38	55.0%	MEDIUM
23	2025	52	49	64.0%	MEDIUM
24	2025	71	74	72.0%	HIGH
25	2025	98	103	81.0%	HIGH

26	2025	0	142	78.0%	HIGH
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Methodology & Model Notes

Model Architecture & Methodology: The FedXGNN Explainable Federated Learning Framework employs a split Graph Attention Network (GAT) trained on multi-modal environmental sensor streams fused with anonymised EHR records. Each hospital edge node processes patient data locally via spaCy NLP, then transmits a 64-dimensional temporal embedding to the central GAT which aggregates spatial outbreak risk across 284 Indian districts. SHAP (KernelSHAP) values are computed per inference against a 15-sample historical baseline to provide XAI-grade attribution. Outbreak probability thresholds — HIGH (>70%), MEDIUM (>40%), LOW (≤40%) — are calibrated against WHO Dengue surveillance benchmarks. Raw patient data never leaves the edge node, ensuring full privacy compliance. All outputs should be reviewed by a qualified epidemiologist before clinical or policy action.